

Antoine Boucher

SOFTWARE ENGINEER — PLATFORM AND COMPUTER GRAPHICS

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Summary

Cloud-native software engineer focused on platform engineering and computer graphics. Ship Kubernetes, GCP, Terraform, GitLab CI/CD, and observability for Java/Kotlin/Spring Boot and Python microservices; harden releases with JUnit, Playwright, and DevSecOps practices. Master's at ÉTS with real-time rendering, physics simulation, and tactile sensing in C++, OpenGL/GLSL, and Blender.

Skills

Graphics / 3D	Computer Graphics, OpenGL, GLSL, Real-time rendering, Physics simulation, C++
Graphics / 3D (intermediate)	Blender, CUDA, OpenCV
Backend	Java, Kotlin, Spring Boot, Python, Django, Flask, REST APIs, Microservices
Backend (intermediate)	C#, .NET, Android, GraphQL
Frontend	JavaScript, HTML, Angular, React
Frontend (intermediate)	Vue, CSS
CI/CD & Pipelines	DevOps, DevSecOps, Git, GitLab CI/CD, GitLab Pipelines, GitHub Actions
CI/CD & Pipelines (intermediate)	Azure DevOps
Cloud / Platform	Docker, Kubernetes, Docker Swarm, IaC, GCP, Azure, AWS
Cloud / Platform (intermediate)	Terraform
Data / Storage	SQL, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Data Science
Testing / Quality	Playwright, Selenium, JUnit
Observability	ELK, Elasticsearch, Kibana, Grafana, Prometheus
Observability (intermediate)	Sentry, Serilog
Other	Linux, Bash, Ansible, Auth0, OWASP, Secure coding, SCRUM, SFAE, MCP, Business Research
Credentials	AWS Certified Cloud Practitioner; QCES; AWS Cloud Quest

Experience

IMC2 – Institut multidisciplinaire en cybersécurité et cyberrésilience

Montréal, QC (Remote)

Research Assistant — Secure cloud platforms (Permanent Part-time)

Mar 2025 – Feb 2026

- Owned 5 GitLab CI/CD pipelines (split a monolith repo into per-service micro-repos) and runners across two GKE clusters on GCP, with Terraform-managed networking and observability, tightening repeatable cloud releases
- Containerized workloads in Docker; configured and operated the research GKE clusters on GCP (workload networking); rolled out services with Terraform, making infrastructure changes safer and faster for the team
- Hardened 5 Java microservices with JUnit and Playwright; promoted Android builds through GitLab into the Terraform-managed GKE environment, raising pre-release coverage and syncing mobile with backend delivery
- Shipped scripted cybersecurity risk analysis across 1000+ risks and 181+ threat categories with weighted scoring, star-based prioritization, and an Angular triage UI so analysts triage faster with consistent context
- Technical Skills:** Java, Kotlin, Android, Docker, Kubernetes, Terraform, GitLab CI/CD, GCP, JUnit, Playwright, Angular

Polytechnique Montréal

Montréal, QC (On-site)

Teaching Assistant — DevSecOps (LOG 8100) (Contract Part-time)

Sep 2024 – Jan 2026

- Built and maintained the GitLab workspace (repos, CI/CD templates, runners, workflows) for approximately 20 concurrent teams, reducing admin friction during the term
- Coached Docker and Kubernetes build, deploy, and debug under LOG 8100 goals; anchored security with OWASP-driven examples, connecting lecture theory to exploitable scenarios
- Refreshed DevSecOps-aligned lab artifacts, graded practicals, and coached resubmissions, improving submission quality and turnaround
- Technical Skills:** GitLab, CI/CD, DevSecOps, Docker, Kubernetes, OWASP, Secure Coding, Linux

École de technologie supérieure (ÉTS)

Teaching Assistant (Contract Part-time)

Montréal, QC (Hybrid)

Feb 2024 – Apr 2026

- Delivered labs, integrator-project support, and office hours on mobile/UX, integrator-project, and distributed-data tracks, totaling approximately 150 contact hours (e.g., TCH 057, IND 500, TCH 099, GTI 660, GTI 320)
- Authored and refreshed SQL, MongoDB, Kafka, and PostgreSQL exercises with rubrics for IND 500, GTI 660, and related cohorts, matching assessments to learning outcomes
- Unblocked integrator teams on architecture and backend/data; sustained Azure-backed course infrastructure and SCRUM coaching; kept mobile/UX and GTI 320 sessions on track, so students shipped end-to-end features
- **Technical Skills:** SQL, MongoDB, Kafka, PostgreSQL, Distributed Databases, Microsoft Azure, Android Studio, UI/UX, SCRUM

IONODES

Cloud Developer (Permanent Full-time)

Laval, QC (Hybrid)

May 2023 – Aug 2023

- Designed a three-tier Auth0 subscription model for organization-scoped IoT access, gating product capabilities per tenant
- Shipped Sentry SDK integrations in C# with Serilog for structured production errors, improving diagnosability for support and engineering
- Deepened ONVIF and WebRTC streaming readiness at IoT scale; orchestrated Azure DevOps with feature flags and repeatable promotions, de-risking releases
- **Technical Skills:** C#, .NET, Azure DevOps, Auth0, Sentry, Serilog, UML, Microservices, ONVIF, WebRTC

Intact

Back End Developer (Co-op)

Montréal, QC (Hybrid)

Jan 2022 – May 2022

- Delivered a 6-step signup flow and mobile-app observability on Kotlin/Spring Boot microservices serving 10,000+ active users, accelerating time-to-active coverage
- Implemented Kotlin login and MFA APIs for client mobile apps while preserving policy controls
- Extended the ELK stack with Kibana dashboards and harmonized microservice error codes, shortening incident triage for operations
- **Technical Skills:** Kotlin, Spring Boot, GraphQL, MongoDB, Microservices, Docker Swarm, Jira, Kibana, SAFe

Divine Essence (Union-Nature)

Web Specialist (Contract Part-time)

Dorval, QC (On-site)

Jun 2020 – Dec 2020

- Modernized the bilingual CommerceBuild storefront (FR/EN blog navigation, accessibility, SEO sales labels, custom Add-to-Cart API, 3CX chat-bot), lifting qualified traffic and self-serve support
- **Technical Skills:** HTML, CSS, JavaScript, CommerceBuild, 3CX, SEO

Power Go

Full Stack Web Developer (Co-op)

Montréal, QC (Remote)

Sep 2020 – Apr 2021

- Built a Python/Django/Vue.js/PostgreSQL feed parser unifying XML/JSON/CSV/Excel supplier feeds, integrated Facebook Shop and Marketplace APIs, and shipped a CAD/USD FX-rate service tied to inventory dates, keeping cross-border catalogs in sync without double entry
- **Technical Skills:** Python, Django, Vue.js, PostgreSQL, MongoDB, JavaScript, Git, Jira, Full-Stack Development

Wandrian

Full Stack Developer (Co-op → Part-time)

Montréal, QC (On-site)

May 2019 – Jan 2020

- On the Italiarail stack with Railkey Tech, integrated Braintree and extended purchase APIs while rebuilding the Selenium E2E harness and Django/Pyramid/jQuery internal tools, stabilizing bookings and catching UI regressions early
- **Technical Skills:** Python, Django, Pyramid, JavaScript, jQuery, Selenium, MongoDB, Git, Jira, Test Automation

Algolux

Engineering Intern

Montréal, QC (On-site)

May 2017 – Aug 2017; May 2018 – Aug

2018

- Built C++/CUDA/OpenCV imaging tooling for camera/ISP validation (HDR Nexus 6 rigs, Python RAW-color processing, Flask REST capture) and produced an approximately 12,000-image outdoor vision dataset, shrinking manual validation loops
- **Technical Skills:** Python, C++, CUDA, OpenCV, Flask, Ansible, Linux, Windows

Education

École de technologie supérieure (ÉTS)

Master's Degree, Information Technology Engineering

Montréal, QC, Canada

Aug 2023 – Apr 2026

- Thesis: couples GelSight-style tactile sensing with interactive real-time physics so wear updates friction and shifts texture-space coordinates as the surface geometry changes.

- AlgoÉTS (ÉTS algorithmic trading club) — President; club infrastructure, web, and trading/data projects.
- **Academic focus:** software engineering, distributed systems, databases, and team-based integrator delivery through ÉTS labs and coursework.